Midterm Exam

ORF 245 - Fundamentals of Statistics
Fall 2023

Date: Wednesday, October 11, 2023

Instructions:

- Please do not turn over this page until you are told to start the exam.
- Please write all of your solutions in the booklet, and make sure to write your name on the cover of the booklet.
- The exam is 80 minutes (i.e., from 3:00-4:20pm).
- Other than one 8.5x11 inch sheet of notes, this is a closed book exam. So, please do not use any other resources (book, notes, calculator, cell phone, etc.).
- The exam has two parts for a total of 60 points.
- Part I has 20 short answer questions worth one point each. You do **not** need to show any work (other than your answers) for the questions in Part I.
- Part II has four problems worth 10 points each. Please **show your work** for the questions in Part II.
- At the end of the exam, please put your exam inside of the booklet and return BOTH.
- No collaboration allowed. You should not discuss the exam questions with others until after the exam is over.
- Good luck!

Part I (1 point each, 20 points total)
You do **NOT** need to show any work (other than your answers) for this part.

1. True or False? The sample variance is always greater than the population variance (with some fixed $x_1, ..., x_n$).

2. True or False? For any two events, $P(A \cup B) \geq P(A)\,P(B)$.

3. True or False? For any two events, $P(A \cap B) \leq P(A)\,P(B)$.

4. Suppose a PIN has four digits using any of the ten digits $0, 1, \ldots, 9$. How many different possible PINs are there if digits can be repeated?

5. True or False? For any $n > 1$ and any $1 < k \leq n$, the number of *permutations* of size k from n objects is strictly larger than the number of *combinations* of size k from n objects.

6. In a race of 100 runners, how many different ways can the top three medals be handed out?

7. If $P(A) = 1/2$ and $B = A^c$, then what is $P(A|B)$?

8. True or False? Consider tossing a fair die (i.e., outcomes $1, \ldots, 6$ all have equal probability), then the events $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$ are independent.

9. If $X \sim \mathrm{Bin}(n, p)$, what is the largest value that X can take on?

10. What is the range of a Poisson random variable?

11. True or False? A Gaussian random variable is appropriate for modeling the number of customers arriving at a store.

12. True or False? The pdf $f(x)$ for a continuous random variable X must satisfy $0 \leq f(x) \leq 1$ for every $x \in \mathbb{R}$.

13. True or False? If a continuous random variable X has pdf $f_X(x)$ and continuous random variable Y has pdf $f_Y(y)$, then $f(x, y) = f_X(x) f_Y(y)$ is a valid joint pdf.

14. True or False? For any two random variables X and Y, $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

15. True or False? If X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$.

16. What is $\mathbb{E}[X^4]$ in terms of $\mathbb{V}[X^2]$ and $\mathbb{E}[X^2]$?

17. Write the pdf of a Gaussian distribution with mean equal to -1 and variance 4.

18. If X has a normal distribution with mean -1 and variance 4, express $P[-5 \leq X \leq 1]$ in terms of the cdf of the standard normal distribution $\Phi(z)$.

19. True or False? For two random variables X and Y, if $\mathbb{C}\mathrm{ov}(X, Y) = 0$ then the correlation $\rho_{X,Y} = 0$.

20. True or False? If the correlation between X and Y is ρ, then the correlation between $-X$ and Y is equal to $-\rho$.

Part II (10 points each, 40 points total)
Please **show your work** for this part.

1. (10 points)
   Let $A, B \subseteq S$. Prove or disprove (i.e. by counterexample) the following:
   (a) $(A \cap B)^c \subseteq A^c \cup B^c$ (Do NOT use DeMorgan's Law).
   (b) $A^c \cup B^c \subseteq (A \cap B)^c$ (Do NOT use DeMorgan's Law).
   (c) $A \cup B^c = B \cup A^c$.
   (d) $A \cap (B \cup B^c) = A$.

2. (10 points)
   Suppose there are two bags with three marbles each. One of the bags (call it Bag 0) has three red marbles and no white marbles, while the other bag (call it Bag 1) has two red marbles and one white marble. One of the two bags is chosen at random (with equal probability). You don't know which bag has been chosen, but you get to pull out two marbles (one at a time) from the bag *without replacement* and observe their colors.
   (a) What is the probability that the first marble you pull out is red?
   (b) What is the probability that first marble and the second marble you pull out are both red?

(c) What is the probability that the bag chosen (i.e., from which you pulled out the marbles) is Bag 0 given that the two marbles you pulled out are both red?

3. (10 points)
Recall that the probability density function of an exponential random variable (parameterized by λ) is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Suppose $X \sim \text{Exp}(\lambda)$, i.e., the pdf of X is as given above.

(a) Find and sketch the cdf of X. Make sure to label your axes.

(b) For a fixed $t \geq 0$, find $P[X \geq t]$.

(c) For a fixed $t \geq 0$, find the conditional density of X given that $X \geq t$. I.e., find $f_{X|X\geq t}(x)$.

(d) What is $\mathbb{E}[X|X \geq t]$? (Hint: you can use the fact that $\mathbb{E}[X] = \frac{1}{\lambda}$.)

(e) Provide a one-sentence interpretation of these results – why might the exponential distribution be called "memoryless"?

4. (10 points)
Suppose (X, Y) are discrete random variables with joint pmf

	$p(x, y)$	y = 0	1	2
	0	0.1	0.1	0.3
x	1	0.1	0.1	0
	2	0.2	0	0.1

(a) Find the marginal pmf's $p_X(x)$ and $p_Y(y)$.

(b) Are X and Y independent? Justify your answer.

(c) Find the conditional distribution $p_{X|Y}(x|y = 2)$.

(d) Find the covariance $\mathbb{C}\text{ov}(X, Y)$.

(e) Find $P(X = 0 \,|\, X + Y = 2)$.